

Engineering Foundation & Requirements Baseline

Software Engineering 381-SEN381
Milestone 1

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Problem Statement

CivicConnect is intended to address challenges associated with managing service requests through fragmented and informal processes. In many organisations, requests submitted through multiple communication channels can result in poor visibility, duplicated records, inconsistent handling procedures, and limited accountability. Research shows that structured service management approaches improve transparency, monitoring, documentation, and organisational control compared to unstructured service delivery processes (Serrano et al., 2021; Hochstein, Tamm and Brenner, 2005).

The primary stakeholders affected by these problems are service requesters (students/citizens), operational staff who process requests, managers responsible for oversight, and organisational decision-makers who depend on accurate operational information (Sharp, Finkelstein and Galal, 1999; Rahman, Moonira and Zuhora, 2015).

The current process is ineffective because it lacks centralised monitoring, standardised workflows, reporting capabilities, and traceable records of request activities. Studies on service-oriented IT management indicate that organisations benefit from having a single point of contact, standard processes, workflow management, and reporting mechanisms to improve service delivery and operational efficiency (Hochstein, Tamm and Brenner, 2005).

If these issues are not addressed, stakeholders may experience delays in service delivery, reduced transparency, incomplete request tracking, inconsistent prioritisation of requests, and difficulty monitoring organisational performance. Furthermore, poorly managed requirements and processes can contribute to project and operational failures, increased costs, and reduced stakeholder satisfaction (Dekkers and Forselius, 2007; Zhou, 2004).

Business Need

CivicConnect needs to be developed to provide a centralised platform for managing service requests in a structured, transparent, and efficient manner. Service-oriented management research shows that organisations achieve improvements in customer focus, service quality, process control, operational efficiency, and transparency through the implementation of structured service management systems (Hochstein, Tamm and Brenner, 2005; Serrano et al., 2021).

The system will provide value by allowing users to submit requests electronically, track request progress, receive service updates, and interact with a consistent service-delivery process. Functional requirements that support stakeholder goals contribute directly to improved user satisfaction and more effective service delivery (Malan and Bredemeyer, 1999).

From an organisational perspective, CivicConnect will improve operational efficiency by introducing standardised workflows, process monitoring, reporting mechanisms, request prioritisation, and auditability. Research indicates that standardisation and process automation contribute significantly to efficiency improvements and better organisational performance (Hochstein, Tamm and Brenner, 2005).

The system also supports accountability because all requests can be documented, monitored, tracked, and reported throughout their lifecycle. Detailed process monitoring and reporting are recognised benefits of modern service management systems (Serrano et al., 2021; Hochstein, Tamm and Brenner, 2005).

For users, the system provides transparency, predictable service delivery, easier communication, and the ability to monitor request progress. Service-oriented management approaches have been shown to improve customer orientation and service quality by focusing processes around stakeholder needs (Hochstein, Tamm and Brenner, 2005).

Business Value Statement

CivicConnect will deliver business value by centralising service-request management, improving service quality, increasing transparency and accountability, standardising operational processes, and providing stakeholders with accurate information for decision-making and service delivery improvement (Hochstein, Tamm and Brenner, 2005; Serrano et al., 2021).

2. Stakeholder Analysis

Stakeholders are individuals, groups, or organisations that are affected by a system or that can influence its requirements either directly or indirectly (Sharp, Finkelstein and Galal, 1999). Successful requirements engineering depends on identifying and consulting the appropriate stakeholders because their input influences the quality, completeness, and success of the resulting system (Sharp, Finkelstein and Galal, 1999; Rahman, Moonira and Zuhora, 2015).

Stakeholder analysis involves identifying stakeholders, understanding their needs, evaluating their influence, and recognising potential conflicts between stakeholder groups. This process helps ensure that system requirements address the needs of all relevant parties and reduces the risk of missing critical requirements (Rahman, Moonira and Zuhora, 2015).

Stakeholder Identification

The CivicConnect system affects multiple stakeholder groups because it serves as a platform for submitting, managing, monitoring, and reporting on service requests.

Who is affected by the system?

The stakeholders affected by CivicConnect include:

- Requesters (citizens who submit service requests)
- Staff members responsible for processing and resolving requests
- Department managers who monitor service delivery performance and operational outcomes
- System administrators responsible for maintaining system functionality and user accounts
- IT support personnel responsible for technical maintenance and support
- Municipal or organisational management who sponsor and own the system
- Regulatory and compliance stakeholders who establish legal and privacy requirements that affect the system

Stakeholder Register

Stakeholder	Role	Needs	Interest	Influence
Requesters	Submit service requests	Simple request submission, status tracking, notifications, transparency	High	Medium
Staff Members	Process requests	Request assignment, prioritisation, status updates, workflow support	High	High
Department Managers	Monitor operations	Reporting, monitoring, oversight, performance information	High	High
System Administrator	System management	User management, permissions, system configuration	High	High
IT Support	Technical support	System maintenance, troubleshooting, reliability	Medium	Medium

Municipal Management / Sponsors	Project ownership	Improved service delivery, efficiency, accountability	High	High
Compliance / Regulatory Stakeholders	Governance and compliance	Privacy protection, security, legal compliance	Medium	High

The stakeholder categories above align with recognised stakeholder classifications such as users, operators, decision-makers, sponsors, and regulators identified in stakeholder-analysis research (Sharp, Finkelstein and Galal, 1999; Rahman, Moonira and Zuhora, 2015).

Stakeholder Needs

Requesters

Requesters require:

- A simple method for submitting service requests
- The ability to track request progress and current status
- Visibility and transparency throughout the request lifecycle
- Reliable communication regarding request updates and outcomes

Staff Members

Staff members require:

- Tools to manage assigned requests efficiently
- Standardised workflows for request processing
- The ability to prioritise requests appropriately
- Access to request histories and audit information

Department Managers

Managers require:

- Reporting and performance-monitoring capabilities
- Visibility into request volumes, status distributions, and service performance
- Accountability and audit mechanisms to support oversight

System Administrator

System administrators require:

- User and role management capabilities
- Access control management
- System configuration functionality
- Audit and monitoring tools

IT Support

IT support personnel require:

- Access to maintenance functionality
- System-health monitoring and troubleshooting capabilities
- Reliable operational information for diagnosing technical issues

Stakeholder Influence

Stakeholders should be analysed according to their interest, power, influence, knowledge, and ability to affect project decisions because these characteristics determine their impact on system requirements (Rahman, Moonira and Zuhora, 2015).

Stakeholder	Can Influence Requirements?	Can Approve/Reject Features?
Requesters	Yes	No
Staff Members	Yes	Limited
Department Managers	Yes	Yes
System Administrator	Yes	Limited
IT Support	Yes	Limited
Municipal Management / Sponsor	Yes	Yes
Compliance Stakeholders	Yes	Yes, for compliance-related functionality

Requirements should be linked directly to stakeholder needs because stakeholder input forms the foundation of requirements engineering activities (Sharp, Finkelstein and Galal, 1999).

Stakeholder Conflicts and Competing Expectations

Research shows that stakeholders often have different goals and priorities, which can create conflicts during requirements engineering and system design (Sharp, Finkelstein and Galal, 1999; Rahman, Moonira and Zuhora, 2015).

Conflict 1: Speed vs Control

Requesters may want requests resolved as quickly as possible, while managers may require approval processes, escalation procedures, and compliance controls that lengthen processing time (Rahman, Moonira and Zuhora, 2015).

Conflict 2: Simplicity vs Detail

Users generally prefer a simple and fast request-submission process, whereas staff and administrators may require additional information for categorisation, auditing, prioritisation, and reporting purposes (Rahman, Moonira and Zuhora, 2015).

Conflict 3: Transparency vs Security

Requesters may expect access to extensive request information and status updates, while administrators and compliance stakeholders must restrict access to sensitive information to ensure privacy and regulatory compliance (Sharp, Finkelstein and Galal, 1999).

Conflict 4: Operational Efficiency vs Reporting Requirements

Staff members may prioritise quick request processing, while managers require detailed reporting, auditing, and documentation to support accountability and performance measurement (Hochstein, Tamm and Brenner, 2005; Serrano et al., 2021).

3. Scope Baseline

The CivicConnect project will follow a Minimum Viable Product (MVP) approach by prioritising core functionality that provides immediate value to stakeholders while allowing additional functionality to be introduced in future releases. MVP development reduces uncertainty, improves learning about stakeholder needs, and avoids unnecessary development effort during the early stages of a project (Anderson, Lim and Joglekar, 2017).

In-Scope Features

The first release of CivicConnect will focus on the core functionality required to support service-request management and improve transparency, accountability, and operational efficiency (Serrano et al., 2021; Hochstein, Tamm and Brenner, 2005).

User Management

- User registration and account creation
- User login and authentication
- Role-based access for requesters, staff, managers, and administrators

Request Management

- Submit a service request electronically
- View submitted requests
- Update request information
- Categorise requests using predefined categories
- Assign requests to appropriate staff members
- Prioritise requests according to urgency and importance

Request Tracking

- Track request status throughout its lifecycle
- View request history and status updates
- Maintain an audit trail of actions performed on requests

Management and Reporting

- Generate operational reports
- Monitor request trends and service performance
- View request statistics and status distribution reports

Administration

- User account management
- Request category management
- System configuration and maintenance functions

Out-of-Scope Features

The following features are deliberately excluded from the initial release to maintain a manageable project scope and ensure that core functionality is delivered successfully. Research indicates that uncontrolled scope expansion increases project complexity, risk, cost, and schedule pressure (Dekkers and Forselius, 2007). [\[Increase Limit with Concrete | PDF\]](#)

Mobile Application

A dedicated Android or iOS mobile application will not be developed during the initial release

AI Chatbot

Automated conversational support through an AI chatbot will not be implemented during the first release

AI-Based Request Classification

Intelligent request categorisation and automated analysis will not be developed during the initial phase

SMS Integration

SMS notifications and SMS-based request submission will not be included because external service integration increases technical complexity and development effort (Dekkers and Forselius, 2007).

Predictive Analytics

Advanced forecasting, predictive reporting, and machine-learning-based insights are excluded from the current scope.

Future Scope

The CivicConnect architecture should be designed so that future functionality can be incorporated as stakeholder needs evolve. Research supports delivering core functionality first while allowing systems to be expanded through future releases (Malan and Bredemeyer, 1999; Anderson, Lim and Joglekar, 2017).

Potential Future Features

- Mobile application development
- SMS and email notification integration
- AI-assisted request routing and prioritisation
- AI chatbot support services
- Integration with municipal, university, or enterprise systems
- Cloud-based service integration
- Advanced analytics and business intelligence dashboards
- Automated escalation and workflow optimisation

Scope Decisions and Justification

Why Certain Features Were Included

The selected in-scope functionality represents the minimum set of features required to support request submission, request processing, request tracking, reporting, and system administration. Service-management research identifies these capabilities as foundational requirements for improving transparency, operational efficiency, monitoring, and service quality (Serrano et al., 2021; Hochstein, Tamm and Brenner, 2005).

Furthermore, stakeholder-driven requirements engineering recommends prioritising functionality that directly supports stakeholder goals and operational needs (Malan and Bredemeyer, 1999; Rahman, Moonira and Zuhora, 2015).

Deliberate Exclusion and Defence

AI Chatbot Exclusion

The AI chatbot feature has been excluded from the initial release.

Justification:

The MVP approach recommends prioritising essential business functionality before investing resources into advanced features. Request submission, tracking, reporting, and management directly support stakeholder goals and business objectives, whereas chatbot functionality provides additional convenience rather than core business value. Focusing on essential functionality first reduces uncertainty, minimises wasted development effort, and improves project feasibility (Anderson, Lim and Joglekar, 2017).

SMS Integration Exclusion

SMS integration has also been excluded from the initial release.

Justification:

SMS functionality requires third-party services, additional infrastructure, integration effort, and increased project complexity. Scope-management research indicates that expanding project scope introduces additional schedule, budget, and risk implications. Therefore, SMS functionality is deferred to a future release while the initial project focuses on core service-management capabilities (Dekkers and Forselius, 2007).

Scope Statement

The CivicConnect project will deliver a web-based service-request management platform that supports user registration, authentication, request submission, request tracking, request assignment, prioritisation, reporting, auditing, and administrative management. Advanced functionality, including mobile applications, AI-assisted services, SMS integration, predictive analytics, and external system integration, is intentionally excluded from the initial release and reserved for future development phases. This scope supports an MVP strategy that prioritises stakeholder value, reduces project risk, and focuses on delivering the core functionality necessary to improve service transparency, accountability, and operational efficiency (Anderson, Lim and Joglekar, 2017; Dekkers and Forselius, 2007; Serrano et al., 2021).

4. Functional Requirements

Functional requirements describe the services, functions, and behaviours that a system must provide in order to achieve its intended purpose and satisfy stakeholder needs (Zhou, 2004; Malan and Bredemeyer, 1999). Functional requirements define what the system must do and serve as the foundation for design, implementation, testing, and validation activities (Zhou, 2004; Malan and Bredemeyer, 1999). Requirements should be clearly documented and assigned unique identifiers to support traceability throughout the software development lifecycle (Zhou, 2004).

The functional requirements for CivicConnect have been derived from the identified stakeholder needs, business requirements, service-management principles, and project scope baseline (Sharp, Finkelstein and Galal, 1999; Rahman, Moonira and Zuhora, 2015).

Functional Requirements List

User Functions

ID	Functional Requirement	Source
FR001	The system shall allow users to register an account.	Requester Needs; User Management Requirements
FR002	The system shall allow registered users to log into the system.	Requester Needs; User Authentication Requirements

FR003	The system shall allow users to submit a service request electronically.	Business Need; Service Request Management Principles
FR004	The system shall allow users to select a predefined request category when creating a request.	Service Management Workflow Requirements
FR005	The system shall generate a unique request number for every submitted request.	Requirements Traceability Principles
FR006	The system shall allow users to view all requests submitted by them.	Requester Needs; Transparency Requirements
FR007	The system shall allow users to track the status of a submitted request.	Business Need; Process Monitoring Requirements
FR008	The system shall allow users to view the history of updates made to their requests.	Auditability and Monitoring Requirements

Staff Functions

ID	Functional Requirement	Source
FR009	The system shall allow staff members to view requests assigned to them.	Staff Requirements; Workflow Management

FR010	The system shall allow staff members to assign requests to appropriate personnel.	Service Workflow and Assignment Processes
FR011	The system shall allow staff members to update request statuses.	Request Management Requirements
FR012	The system shall allow staff members to prioritise requests according to urgency and importance.	Prioritisation Models in Service Management
FR013	The system shall allow staff members to add comments or resolution notes to requests.	Process Documentation and Monitoring Requirements
FR014	The system shall maintain a record of all actions performed on each request.	Audit Trail and Transparency Requirements

Manager Functions

ID	Functional Requirement	Source
FR015	The system shall allow managers to view operational reports.	Management Reporting Requirements
FR016	The system shall allow managers to view request statistics and trends.	Monitoring and Reporting Requirements

FR017	The system shall allow managers to monitor request performance indicators.	Service Performance Monitoring Requirements
FR018	The system shall allow managers to view overdue or unresolved requests.	Service Oversight Requirements

Administrator Functions

ID	Functional Requirement	Source
FR019	The system shall allow administrators to create, update, and deactivate user accounts.	Administrative Management Requirements
FR020	The system shall allow administrators to assign user roles and permissions.	Stakeholder and Administrative Management Requirements
FR021	The system shall allow administrators to manage request categories.	Administrative Configuration Requirements
FR022	The system shall allow administrators to configure system settings.	System Administration Requirements
FR023	The system shall allow administrators to view system audit logs.	Audit and Monitoring Requirements

What Must the System Do?

The CivicConnect system must:

- Allow users to register and authenticate accounts
- Allow service requests to be submitted electronically
- Track and monitor request progress throughout the request lifecycle
- Support request assignment and prioritisation workflows
- Generate reports for service delivery monitoring and decision-making
- Maintain audit trails and request histories for accountability and transparency
- Support user, role, and configuration management functions

What Functions Must Users Perform?

Users must be able to:

- Register and log into the system
- Submit service requests
- Select request categories
- View submitted requests
- Track request progress and status updates
- View request history information

What Functions Must Staff Perform?

Staff members must be able to:

- View assigned requests
- Assign requests to responsible personnel
- Prioritise requests according to urgency
- Update request statuses
- Record comments and resolutions
- Maintain accurate records of request activities

What Functions Must Administrators Perform?

Administrators must be able to:

- Manage user accounts
- Assign permissions and user roles
- Configure request categories and system settings
- Review audit logs and monitoring information
- Maintain operational integrity of the system

5. Acceptance Criteria

Acceptance criteria define the measurable conditions that must be met before a functional requirement can be considered successfully implemented. Requirements should be verifiable, testable, complete, and capable of validation to ensure that the system satisfies stakeholder needs and business objectives (Zhou, 2004; Malan and Bredemeyer, 1999). Functional requirements must therefore be accompanied by acceptance criteria that enable testers, developers, and stakeholders to determine whether the required functionality has been implemented correctly (Zhou, 2004).

The acceptance criteria below are derived from the functional requirements identified for CivicConnect and are aligned with service-management principles that emphasise transparency, accountability, monitoring, reporting, and process control (Serrano et al., 2021; Hochstein, Tamm and Brenner, 2005).

FR001: User shall register an account.

Acceptance Criteria

- User can access the registration page.
- User can enter all required registration information.
- The system validates mandatory fields.
- The system prevents duplicate accounts.
- The system creates the account successfully.
- A confirmation message is displayed.

Justification: Requirements must be testable and validated against stakeholder needs (Zhou, 2004).

FR002: User shall log into the system.

Acceptance Criteria

- User can access the login page.
- User can enter valid credentials.
- The system authenticates valid users.
- Invalid credentials are rejected.
- Authenticated users are redirected to the dashboard.

Justification: Functional requirements describe services that the system must perform and should be verifiable during testing (Malan and Bredemeyer, 1999; Zhou, 2004).

FR003: User shall submit a service request.

Acceptance Criteria

- User can open the request submission form.
- User can enter all required request information.
- User can select a request category.
- User can submit the request successfully.
- The system stores the request in the database.
- A unique request number is generated.
- A confirmation message is displayed.

Justification: Service-management systems require structured request submission processes and traceable request records (Hochstein, Tamm and Brenner, 2005; Zhou, 2004).

FR004: User shall select a predefined request category.

Acceptance Criteria

- The system displays a list of available categories.
- User can select only one category per request.
- The selected category is stored with the request.
- Requests cannot be submitted without a category.

Justification: Standardised categorisation improves workflow management and request handling (Hochstein, Tamm and Brenner, 2005).

FR005: System shall generate a unique request number.

Acceptance Criteria

- Every submitted request receives a unique identifier.
- Duplicate request numbers cannot exist.
- The request number is visible to the requester.
- The request number can be used when searching for a request.

Justification: Requirements and project artefacts should be traceable through unique identifiers (Zhou, 2004).

FR006: User shall view submitted requests.

Acceptance Criteria

- User can open a request history page.
- The system displays all requests submitted by the user.
- Request information is displayed correctly.
- Users cannot view requests that belong to other users.

Justification: Transparency and visibility are important characteristics of service-management systems (Serrano et al., 2021)

FR007: User shall track request status.

Acceptance Criteria

- User can view submitted requests.
- Current request status is displayed.
- Status updates are displayed immediately after processing.
- User can identify whether a request is pending, assigned, in progress, or completed.

Justification: Monitoring and transparency improve service delivery and accountability (Serrano et al., 2021; Hochstein, Tamm and Brenner, 2005).

FR008: User shall view request history.

Acceptance Criteria

- User can open a request-details page.
- Historical updates are visible.
- Comments and status changes are displayed chronologically.
- The complete request history can be viewed.

Justification: Auditability and process monitoring require maintaining historical records of request activities (Serrano et al., 2021).

Staff Functional Requirements

FR009: Staff shall view assigned requests.

Acceptance Criteria

- Staff members can access an assigned-request list.
- The system displays only requests assigned to the logged-in staff member.
- Request details can be viewed.
- Assigned requests are updated automatically when reassigned.

Justification: Service-management workflows depend on effective assignment and monitoring mechanisms (Hochstein, Tamm and Brenner, 2005).

FR010: Staff shall assign requests.

Acceptance Criteria

- Staff members can select a request.

- Staff members can assign requests to another staff member.
- Assignment information is stored successfully.
- The system records who performed the assignment.

Justification: Assignment workflows contribute to process control and service efficiency (Hochstein, Tamm and Brenner, 2005).

FR011: Staff shall update request status.

Acceptance Criteria

- Staff members can select a request.
- Staff members can change the request status.
- The updated status is saved successfully.
- Users can immediately view the updated status.
- The update is recorded in the audit log.

Justification: Status monitoring improves transparency and enables accountability (Serrano et al., 2021).

FR012: Staff shall prioritise requests.

Acceptance Criteria

- Staff members can assign a priority level.
- Priority options are displayed consistently.
- The selected priority is stored successfully.
- Priority information can be viewed by authorised users.

Justification: Prioritisation models support effective service management and resource allocation (Hochstein, Tamm and Brenner, 2005).

FR013: Staff shall add comments and resolution notes.

Acceptance Criteria

- Staff members can add comments.
- Staff members can record resolution information.
- Comments are linked to the correct request.
- Users can view comments where appropriate.
- Comments are included in request history.

Justification: Process documentation improves transparency and monitoring capabilities (Hochstein, Tamm and Brenner, 2005; Serrano et al., 2021).

FR014: System shall maintain an audit trail.

Acceptance Criteria

- Every status change is recorded.
- Every assignment action is recorded.
- Every update is timestamped.
- The user responsible for each action is recorded.
- Audit information cannot be deleted by standard users.

Justification: Auditing supports accountability, traceability, and operational control (Serrano et al., 2021).

Manager Functional Requirements

FR015: Manager shall view operational reports.

Acceptance Criteria

- Managers can generate reports.
- Reports display current request information.
- Reports can be filtered by category or status.
- Reports load successfully.

Justification: Reporting is a key requirement for service monitoring and governance (Serrano et al., 2021).

FR016: Manager shall view request statistics and trends.

Acceptance Criteria

- Statistical summaries are displayed.
- Request counts are calculated correctly.
- Trend information is generated automatically.
- Managers can view historical information.

Justification: Monitoring request trends supports managerial decision-making (Serrano et al., 2021).

FR017: Manager shall monitor performance indicators.

Acceptance Criteria

- Performance metrics are displayed.
- Open and resolved request counts are visible.
- Resolution-time information is available.

- Metrics update when request data changes.

Justification: Service-performance monitoring improves organisational oversight and accountability (Serrano et al., 2021).

FR018: Manager shall view overdue requests.

Acceptance Criteria

- Overdue requests are highlighted.
- Managers can view all overdue requests.
- Overdue status is calculated correctly.
- Overdue requests can be filtered and sorted.

Justification: Managers require visibility into unresolved service issues to support effective oversight (Hochstein, Tamm and Brenner, 2005).

Administrator Functional Requirements

FR019: Administrator shall manage user accounts.

Acceptance Criteria

- Administrators can create accounts.
- Administrators can update account information.
- Administrators can deactivate accounts.
- Changes are saved successfully.

Justification: Administrative functions support operational control and stakeholder management (Rahman, Moonira and Zuhora, 2015).

FR020: Administrator shall assign user roles and permissions.

Acceptance Criteria

- Administrators can assign roles.
- Administrators can modify permissions.
- Users receive permissions associated with their role.
- Permission changes take effect immediately after saving.

Justification: Different stakeholders require different levels of system access and control (Rahman, Moonira and Zuhora, 2015).

FR021: Administrator shall manage request categories.

Acceptance Criteria

- Administrators can create categories.
- Administrators can update categories.
- Administrators can deactivate categories.
- Category changes are visible throughout the system.

Justification: Standardised categories improve workflow management and request organisation (Hochstein, Tamm and Brenner, 2005).

FR022: Administrator shall configure system settings.

Acceptance Criteria

- Administrators can access configuration settings.
- Configuration changes can be saved successfully.
- Updated settings are applied correctly.
- Unauthorised users cannot modify settings.

Justification: System administration capabilities are required for maintaining system operation (Rahman, Moonira and Zuhora, 2015).

FR023: Administrator shall view audit logs.

Acceptance Criteria

- Administrators can access audit records.
- Audit entries display date, time, and action performed.
- Audit logs can be searched.
- Audit data cannot be edited by unauthorised users.

Justification: Audit logging supports transparency, accountability, monitoring, and governance (Serrano et al., 2021).

Traceability, Risk & Forward Planning Lead

4.2 Non-Functional Requirements

These NFRs define measurable quality expectations for CivicConnect. Targets marked as proposed should be validated by the team before baseline sign-off.

ID	Category	Non-Functional Requirement	Source / Stakeholder	Priority	Acceptance / Basis
NFR001	Performance	Normal authenticated pages and request-list operations shall return within 3 seconds under the agreed M1 test load.	Requesters; Staff; Managers	High	Proposed target; validate test environment/load.
NFR002	Performance	Service-request submission shall complete and provide confirmation within 3 seconds under the agreed M1 test load.	Requesters	High	Proposed target; validate with testing.
NFR003	Security	Authentication shall be required before access to protected CivicConnect functions.	All users	High	Verify through access-control tests.
NFR004	Security	Role-based access shall restrict functions and request information to what the user's role is authorised to access.	Requesters; Staff; Managers; Administrators	High	Trace to FR006, FR019–FR023 and security tests.
NFR005	Security	Standard users shall not be able to delete or alter audit records.	Administrators; Compliance	High	Verify with authorisation/security tests.
NFR006	Security	Sensitive request information shall be accessible only to authorised users and shall not be exposed through unauthorised views.	Requesters; Staff; Managers; Compliance	High	Exact sensitive-data categories to be confirmed.
NFR007	Usability	A requester shall be able to submit a valid service request using the defined workflow without requiring assistance.	Requesters	High	Validate through usability acceptance testing.

NFR008	Usability	Validation feedback shall identify missing or invalid mandatory request information before submission is accepted.	Requesters	Medium	Trace to FR003/FR004.
NFR009	Reliability	A successfully submitted request shall receive a unique request number and remain retrievable after submission.	Requesters; Staff	High	Trace to FR003, FR005, FR006.
NFR010	Reliability	A successful status or assignment update shall be persisted and reflected when the authorised request is subsequently viewed.	Staff; Requesters	High	Trace to FR007, FR009–FR011.
NFR011	Auditability	Request status changes, assignments and relevant updates shall record the responsible user and timestamp.	Staff; Managers; Administrators; Compliance	High	Trace to FR011, FR014, FR023.
NFR012	Maintainability	System configuration, user roles and request categories shall be manageable through defined administrative functions.	Administrators; IT Support	Medium	Trace to FR019–FR022.
NFR013	Availability	The agreed test/development deployment shall be available for scheduled demonstrations and verification, excluding planned maintenance.	All stakeholders	Medium	Exact availability target not yet established.
NFR014	Recovery	A recovery approach for CivicConnect request records shall be defined and verified before release evidence is accepted.	Managers; Administrators; IT Support	High	Detailed mechanism deferred to later lifecycle.

6. Requirements Traceability Matrix

The initial RTM links stakeholder/source information to functional requirements and acceptance evidence. Design, implementation, test and release references remain open for later milestones.

Req. ID	Stakeholder / Source	Requirement	Acceptance Evidence
FR001	Requester	Register an account	Registration page; required fields validated; duplicates prevented; account created; confirmation displayed.
FR002	Requester	Log into system	Valid credentials authenticate; invalid credentials rejected; authenticated user reaches dashboard.
FR003	Requester	Submit service request	Required information entered; category selected; request submitted; unique number generated; confirmation displayed.
FR004	Requester	Select predefined request category	Categories displayed; category selected and stored; submission blocked without category.
FR005	Requester / Staff	Generate unique request number	Every submitted request receives a unique identifier; duplicates prevented; number visible/searchable.
FR006	Requester	View submitted requests	User can view own history; information correct; other users' requests inaccessible.
FR007	Requester / Staff	Track request status	Current status displayed; agreed lifecycle states identifiable; updates reflected after processing.
FR008	Requester	View request history	Historical updates, comments and status changes displayed chronologically.
FR009	Staff	View assigned requests	Assigned-request list available; only authorised assigned requests displayed; details accessible.
FR010	Staff	Assign requests	Staff can assign a request; assignment stored; assigning user recorded.
FR011	Staff / Requester	Update request status	Status changed and saved; requester can view update; audit record created.
FR012	Staff	Prioritise requests	Priority selected from consistent options; stored and visible to authorised users.
FR013	Staff / Requester	Add comments and resolution notes	Comments/resolution information linked to correct request and included in history.
FR014	Staff / Admin / Compliance	Maintain audit trail	Status/assignment/update actions recorded with timestamp and responsible user; standard users cannot delete audit information.

FR015	Manager	View operational reports	Manager can access operational reporting.
FR016	Manager	View request statistics and trends	Statistics/trends available to authorised managers.
FR017	Manager	Monitor performance indicators	Defined service-performance indicators available to authorised managers.
FR018	Manager	View overdue or unresolved requests	Manager can identify overdue/unresolved requests.
FR019	Admin	Create, update and deactivate accounts	Administrator can perform defined account-management actions.
FR020	Admin	Assign roles and permissions	Administrator can assign defined roles/permissions.
FR021	Admin	Manage request categories	Administrator can manage defined categories.
FR022	Admin	Configure system settings	Administrator can modify approved settings.
FR023	Admin / Compliance	View system audit logs	Administrator can access audit logs and view recorded system actions.

7. Risk Register

ID	Risk	Cause / Event	Prob.	Impact	Priority	Mitigation	Contingency	Owner	Status
RISK 001	Scope creep	Additional features/integrations expand obligations beyond the controlled MVP.	Medium	High	High	Maintain explicit in/out/deferred scope; require change impact analysis.	Defer/reject non-essential features; update affected artefacts if approved.	Task 2 Lead	Open
RISK 002	Security/privacy failure	Requests may contain sensitive information and access must differ by role.	Medium	High	High	Define security NFRs early; role-based access; least privilege; security tests.	Restrict affected access, investigate, remediate and record residual risk.	Task 2 Lead	Open
RISK 003	Technology learning curve	Unfamiliar technology/platform consumes schedule and reduces testing time.	Medium	High	High	Evaluate capability, compatibility, security, maintainability, testing, deployment and cost.	Reduce non-essential scope; revise plan through controlled change.	Task 2 Lead	Open
RISK 004	Requirement change after baseline	Changes can affect requirements, design, testing, schedule, cost and risk.	Medium	High	High	Use controlled change process and maintain RTM linkage.	Impact-analyse and approve, defer or reject; update affected artefacts.	Task 2 Lead	Open
RISK 005	Deployment/platform	Required technologies/	Medium	High	High	Check compatibility	Select compatibl	Task 2	Open

	incompatibility	services may be unavailable or incompatible in the target environment.				early and record deployment assumptions/dependencies.	alternative or revise scope through controlled change.	Lead	
RISK 006	Insufficient verification evidence	Requirements may lack corresponding tests or acceptance evidence.	Medium	High	High	Use measurable criteria and maintain RTM through testing.	Prioritise high-risk tests and hold release evidence until gaps are closed/accepted.	Task 2 Lead	Open
RISK 007	Data loss/recovery weakness	CivicConnect is intended to become a controlled lifecycle record for requests.	Low/Medium	High	High	Define backup/recovery expectations and verify recovery before release.	Restore from verified backup where possible; document impact/residual risk.	Task 2 Lead	Open
RISK 008	AI-generated content error	AI assistance could introduce inaccurate requirements, citations or engineering claims.	Medium	Medium	Medium	Human verification; log material AI use; do not treat AI output as a source.	Correct affected artefacts, re-review evidence and update AI record.	Task 3 Lead	Open

Highest-priority risk: RISK001 (scope creep) is prioritised because the project has a three-person team, a fixed milestone schedule and explicit scope-control expectations. Uncontrolled additions can create ripple effects across requirements, design, testing, security, deployment, cost and schedule.

8. Forward Engineering Considerations

ID	Concern	Why it matters now	Later decision/activity	Missing information	Risk if ignored
FE001	Security and access control	Role separation and sensitive request information affect requirements now.	Define/validate authentication, authorisation, least privilege and security tests in later design/construction.	Exact sensitive-data categories and final access rules.	Unauthorised disclosure, incorrect permissions and late security redesign.
FE002	Testability and quality evidence	M1 requirements must be measurable so later testing can prove acceptance.	Map requirements to test cases, risk-based testing and release evidence.	Final test strategy, environments and quality thresholds.	Unverifiable requirements and weak defence evidence.
FE003	Deployment and environment compatibility	Technology/platform is not prescribed and compatibility can constrain delivery.	Assess target environment, deployment, configuration, secrets and rollback.	Final hosting/deployment environment and platform constraints.	Late incompatibility, deployment delay or forced scope reduction.

FE004	Data recovery and operational resilience	CivicConnect is intended to provide a controlled lifecycle record.	Define backup, recovery, retention and recovery verification activities.	Recovery objectives, infrastructure and retention requirements.	Loss of request history and weak operational readiness.
FE005	Scalability and performance	Increasing request volume/concurrency can affect response time and platform suitability.	Validate performance assumptions and conduct appropriate performance testing.	Expected user population, workload and capacity.	Poor responsiveness or costly redesign.
FE006	Maintainability and technology evolution	The solution must avoid an unsustainable technical/operational burden.	Evaluate technology maturity, capability, dependency health, maintainability and upgrade path.	Final stack, dependencies and maintenance model.	Technical debt, difficult upgrades and increased lifecycle cost.
FE007	Observability and operations	A live request lifecycle requires detection of failures and operational accountability.	Define logging, monitoring, health checks, failure detection and incident response.	Operational environment and monitoring capabilities.	Failures remain undetected and accountability is weakened.

9. Engineering Decision Log

ID	Decision area	Context	Constraints	Decision / Status	Rationale	Trade-offs / Risks	Later consequence
EDL001	Final technology stack	M1 requirements/constraints exist, but no stack is prescribed and evidence is insufficient for a final selection.	Team capability, compatibility, security, maintainability, testing, deployment and cost.	Defer final selection to M2; evaluate candidates and record an ADR.	Avoids premature commitment while preserving a traceable selection process.	Poor choice could increase learning, security, deployment or maintenance risk.	M2 technology evaluation / ADR
EDL002	Final architecture	M1 establishes business capabilities and quality expectations but not enough design evidence for final architecture.	M1 excludes final architecture.	Defer architecture to M2 and compare options against baselined requirements/NFRs.	Allows architecture to be derived from evidence.	Late architecture choices can affect data, API, UI, security and testing.	M2 architecture ADR
EDL003	External system integrations	External university/municipal/enterprise integrations are outside the initial scope.	MVP scope discipline; schedule/resources.	Defer; reassess only through stakeholder-approved change.	Protects the core request-management objective.	Integration adds security, dependency, testing and schedule risk.	Future change request
EDL004	AI chatbot / AI classification	AI chatbot and AI-based request classification are excluded from initial release.	MVP scope, complexity and risk control.	Defer; reassess future value, cost, security and evidence.	Keeps baseline focused on controlled core	Adds accuracy, privacy, testing and	Future change request

					function ality.	operatio nal obligatio ns.	
EDL 005	Mobile applicati on	Dedicated Android/iOS application is excluded from initial release.	MVP scope and schedule/res ources.	Defer; reassess after core service- request lifecycle is validated.	Avoids duplicate platform work before core value is proven.	Adds platform support and testing effort.	Future change request

Baseline Validation Items

Confirm final stakeholder IDs/naming convention before baseline sign-off.

Confirm which acceptance-criteria details are intended as baseline requirements versus implementation/design assumptions (for example exact status values and database persistence).

Validate proposed NFR targets, especially response-time thresholds and availability expectations, against the available test environment and schedule.

Populate design/architecture, issue/PR, implementation, test and release evidence in the RTM as later milestones produce evidence.

Ensure post-baseline changes follow the controlled change process and update affected requirements, acceptance criteria, RTM, risks and decisions.

Traceability Example for Defence

Requester need for visibility/transparency → FR007 (track request status) → acceptance criteria requiring current status to be displayed and lifecycle states to be identifiable → later design/implementation → test verifying status display → acceptance/release evidence.

Design, implementation and test links are intentionally open until later milestones.

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